

Concept Note — Fund for Innovation in Development (FID)

Stage requested: Stage 0 / Preparation grant (up to €50,000)

Project title: FedFraudShield — Privacy-preserving, collaborative fraud detection to protect the savings of Uganda's SACCO and microfinance members

Country: Uganda (ODA-eligible, OECD DAC list)

Applicant: FraudShield Uganda (*legal entity name and registration number to be inserted*)

Lead: Raymond R. Wayesu, Founder & Lead Data Scientist — raymondwayesu@gmail.com · +256 784 902 753

Proposed research partner: (*to be confirmed — see Section 8*)

Duration: 9 months

Date: September 2026

Status of this document: working draft. Items marked [TBC] must be confirmed or replaced before submission. All FID rules (stage definitions, ceilings, eligibility) must be re-checked against the current call guide on fundinnovation.dev, which is updated periodically.

1. Summary

Uganda's 28,500+ Savings and Credit Cooperative Organisations (SACCOs) and 150+ licensed microfinance institutions (MFIs) hold the savings of roughly 18 million people, most of them rural and low-income. Internal fraud — ghost loans, officer self-lending, collusion rings — erodes an estimated 2–5% of portfolio value each year (UGX 100–250 billion). When a SACCO collapses, its members lose their savings outright; the 2021 PROFIRA assessment found 312 of 453 monitored SACCOs struggling due to fraud and governance failures, and 64 collapsed entirely.

Machine-learning fraud detection is standard in banks but absent from this sector, because no single SACCO has enough data to train a model and Uganda's Data Protection and Privacy Act (DPPA 2019) restricts pooling raw records. **FedFraudShield** is a federated-learning framework that lets many institutions train one shared fraud model without any raw data leaving their premises. In simulation (DSA 2026 workshop paper, in [docs/](#)), federated training raised detection quality (AUPRC) from 0.64 to 0.95 — a 47% relative improvement — and gave institutions with *no* fraud history a working detector. A companion tool, the **FraudShield MVP**, already runs rule-based and statistical checks on any SACCO's exported spreadsheet and is ready for field use.

This Stage 0 grant will convert a validated design into a pilot-ready intervention: secure data-sharing agreements with 3–5 SACCOs, validate the system on real Ugandan data, establish baseline fraud-loss measurements, recruit a research partner, and design the counterfactual evaluation for a Stage 1 pilot.

2. The problem, framed as a poverty and inequality problem

Dimension	Evidence
Who is affected	~18 million SACCO/MFI members; predominantly rural, informal-sector households. Women make up the majority of members in many SACCOs and village savings groups [TBC: cite UCSCU/UMRA figure] .
How they are harmed	Fraud losses are absorbed by members as (a) lost savings when a SACCO fails, (b) higher interest rates and stricter lending to cover losses, (c) loss of trust that pushes households back to informal, higher-cost credit.
Scale of loss	UGX 100–250 billion / year ($\approx$ USD 27–68 million), 2–5% of a >UGX 5 trillion asset base.
Why existing responses fail	Manual audits sample 5–10% of loans; external audits detect ~3% of fraud (ACFE 2024). Commercial banking fraud tools are priced for banks and require data volumes SACCOs do not have.
Why now	SACCOs are digitising rapidly (Ensibuuko, Kanza, Excel), creating the transaction records a detector needs; UMRA's licensing regime and the 2022 Credit Reference Bureau Regulations create both the regulatory pull and a precedent for cross-institution data cooperation.

The intervention is therefore best understood as **protecting the financial assets of the poor** and **keeping community financial institutions solvent**, not as a technology sale to institutions.

3. The innovation

3.1 Two components, one intervention

Component	What it is	Maturity
FraudShield MVP	Browser/Python tool that ingests any SACCO's CSV/Excel export, auto-maps its columns (schema matching), and runs rule-based + statistical detection: duplicate-phone ghost loans, same-day loan stacking, officer self-lending, after-hours approvals, amount outliers, officer concentration.	Working prototype; deployed at <code>mvp/</code> and <code>mvp-fast/</code> in this repository. Ready for real-world testing (FID Stage 1 criterion).
FedFraudShield	Hub-and-spoke federated learning: each SACCO trains a small neural model locally; only differentially-private parameter updates (~100 KB per round) go to an aggregation server hosted in Uganda; the improved global model is returned. No raw data leaves any institution.	Architecture and simulation results published (DSA 2026 workshop, work-in-progress). Not yet run on real Ugandan data.

3.2 What is new

- First application of federated learning to SACCO/MFI fraud detection in Africa (no prior work found in the literature review in the paper).
- Solves the two blockers that have kept ML out of this sector simultaneously: **data scarcity** (pooling learning across institutions) and **privacy law** (DPPA 2019 §14 data minimality; §7(2)(b)(iii) fraud-prevention basis; §19 no cross-border transfer).
- Designed for the actual operating environment: works over 3G (30 MB total for 20 training rounds across 15 institutions), tolerates heterogeneous software and column names, and gives *zero-history*

institutions a usable model from day one.

3.3 Evidence to date (simulation, PaySim dataset, 15 simulated institutions, non-IID)

Method	AUPRC	Recall
Local-only (average institution)	0.643	0.768
Local-only (worst institution)	0.005	0.033
FedAvg (federated)	0.945	0.991
FedProx + differential privacy ($\epsilon = 5$)	0.692	0.312
Centralised (oracle upper bound)	0.977	0.995

Five simulated institutions had no fraud labels at all and could not train any detector alone; all five obtained a working detector through federation.

Known limitations (stated honestly): results are on synthetic mobile-money data, not real SACCO loan books; precision is low at default thresholds (many false positives — acceptable for an investigation-triage tool but must be tuned); strong privacy settings ($\epsilon = 1$) degrade performance significantly. Stage 0 exists to address the first two.

4. Stage 0 objectives and activities

Objective: Reach the point where a Stage 1 pilot with a credible counterfactual design can begin.

#	Activity	Output	Months
A1	Institutional partnerships: sign data-sharing and pilot MoUs with 3–5 SACCOs/MFIs of varied size and software (Ensibuuko, Kanzu, Excel). Engage UMRA and UCSCU as observers.	Signed MoUs; DPPA-compliant data-processing agreements reviewed by counsel.	1–3
A2	Real-data validation: run the FraudShield MVP on 12–24 months of historical records from each partner; validate the schema-mapping module across heterogeneous exports; retrospectively check flagged cases against each SACCO's own audit findings.	Precision/recall on real data; list of fraud typologies present in Ugandan loan data that PaySim lacks (ghost loans, collusion rings).	2–6
A3	Baseline measurement: establish, per partner, current fraud-loss rate, portfolio-at-risk (PAR30/90), write-offs, audit detection rate, and cost of fraud investigation.	Baseline dataset and measurement protocol — the outcome variables for Stage 1/2.	2–6
A4	Federated proof-of-concept on real data: deploy local training nodes at partners, run federated rounds over their actual connectivity, tune thresholds and privacy budget.	Technical feasibility report: bandwidth, latency, model performance vs. local-only on real data.	4–8
A5	Research partnership and evaluation design: formalise an academic partner; pre-register a Stage 1 pilot design (see §5); power calculation from A3 baselines.	Evaluation pre-analysis plan; Stage 1 application.	3–9

#	Activity	Output	Months
A6	Cost model: unit cost per institution per year for both components; comparison with cost of manual audit and average loss averted.	Cost-effectiveness note (\$6).	6–9
A7	Gender-disaggregated analysis: characterise whether women members and women-led SACCOs are differentially exposed to fraud losses.	Short analytic note feeding Stage 1 outcome selection.	5–9

Stage 0 deliverable: a Stage 1 proposal with real-data evidence, signed partners, a pre-registered evaluation design, and a costed budget.

5. Evaluation pathway (what Stage 1 and Stage 2 would test)

FID requires impact evaluation with a counterfactual. The proposed design, to be finalised with the research partner in Stage 0:

- **Unit of randomisation:** SACCO/MFI (the tool operates at institution level).
- **Stage 1 pilot ($\leq$ €200k, ~40–60 institutions):** institutions randomised to (T1) FraudShield MVP alone, (T2) FraudShield + FedFraudShield federation, or (C) waitlist control receiving the tool after 12 months. A phased roll-out design ensures every participant eventually receives the intervention.
- **Primary outcomes:** fraud losses as % of portfolio; PAR30/PAR90; number and value of confirmed fraud cases detected; institution solvency/continuation.
- **Member-level outcomes (Stage 2):** savings retained, access to credit, interest rates charged, member trust and continued participation; disaggregated by gender and rural/urban.
- **Secondary / mechanism outcomes:** time-to-detection, audit cost, staff behaviour (e.g. reduction in after-hours approvals — a deterrence effect that may appear even before a case is proven).
- **Stage 2 ($\leq$ €1.5M):** scale to 150–300 institutions across regions, with a full randomised evaluation powered on member-level outcomes, run by the research partner.

Power will be calculated from Stage 0 baselines (A3). A cluster-randomised design at institution level with ~40–60 clusters is expected to be adequate for institution-level fraud-loss outcomes given the high baseline variance reported by PROFIRA; member-level outcomes will require Stage 2 scale.

6. Cost-effectiveness logic (to be quantified in Stage 0)

Item	Working estimate	Source / status
Annual fraud loss, median SACCO	2–5% of portfolio	Sector estimates; to be replaced by A3 baseline
Cost of FraudShield MVP per institution/year	UGX 6 million ($\approx$ USD 1,600) at current pricing; marginal cost far lower at scale	Current price list
Marginal cost of federation node	Commodity laptop/VM + ~30 MB/month data	Paper \$4.3

Item	Working estimate	Source / status
Break-even	A SACCO with a UGX 500 million portfolio losing 3% (UGX 15 million/yr) breaks even if the tool prevents ~40% of that loss	Illustrative
Comparator	Manual audit covering 5–10% of loans; external audit detecting ~3% of fraud	ACFE 2024

Stage 0 will produce actual cost per institution, cost per fraud case detected, and cost per UGX of loss averted, benchmarked against manual audit.

7. Pathway to scale and sustainability

- Sector infrastructure, not a product:** the federated aggregation hub is designed to be operated as shared infrastructure by a sector body (UMRA as regulator, or UCSCU as the apex cooperative union), analogous to how the Credit Reference Bureau operates. This is the target for FID's *Public Policy Transformation* stage.
- Open standards:** integration with Tazama (Linux Foundation open-source fraud monitoring for emerging markets) so that the model layer is portable and not locked to one vendor.
- Revenue model for the tool layer:** subscription pricing already validated with early conversations; a tiered/free tier for the smallest SACCOs cross-subsidised by MFIs and banks.
- Regional replication:** Kenya, Tanzania and Rwanda have comparable SACCO sectors and data-protection regimes; the federated design transfers directly.

8. Team and partners

Role	Who	Status
Lead / PI	Raymond R. Wayesu — MSc Statistics (Linköping University); Data Analytics Lead, Uganda Virus Research Institute; 10+ years in statistical analysis and ML; author of the FedFraudShield paper	Confirmed
Legal applicant	FraudShield Uganda [TBC: registered entity type, registration no., bank account]	Must be confirmed — FID does not fund individuals
Research partner (evaluation design & Stage 2 evaluation)	Candidates: Makerere University Dept. of Computer Science (Azamuke, Katarahweire, Bainomugisha — cited in the paper); a J-PAL-affiliated economist; Innovations for Poverty Action (IPA) Uganda	[TBC] — to be secured under A5
Implementing partners	3–5 SACCOs/MFIs [TBC: names]	Under discussion
Sector / regulatory observers	Uganda Microfinance Regulatory Authority (UMRA); Uganda Cooperative Savings and Credit Union (UCSCU); Ministry of Trade, Industry and Cooperatives	[TBC]

Role	Who	Status
Legal counsel (DPPA compliance)	[TBC]	Budgeted

9. Budget (Stage 0, 9 months) — indicative, EUR

Line	Amount	Notes
Personnel: lead data scientist (0.6 FTE), one ML engineer (1.0 FTE), one field/partnerships coordinator (0.5 FTE)	24,000	Local salaries
Research partner engagement: evaluation design, power calculations, pre-registration	8,000	Sub-grant or consultancy
Partner SACCO costs: data extraction time, on-site node hardware (5 × laptop/VM), connectivity	6,000	
Legal: DPPA compliance review, data-processing agreements, MoU templates	3,000	
Aggregation server hosting in Uganda + security review	2,500	
Travel and field work (Kampala, Jinja, Mbarara, Gulu, Mbale)	3,000	
Stakeholder workshop (UMRA, UCSCU, partners) and dissemination	1,500	
Contingency (≈4%)	2,000	
Total	50,000	

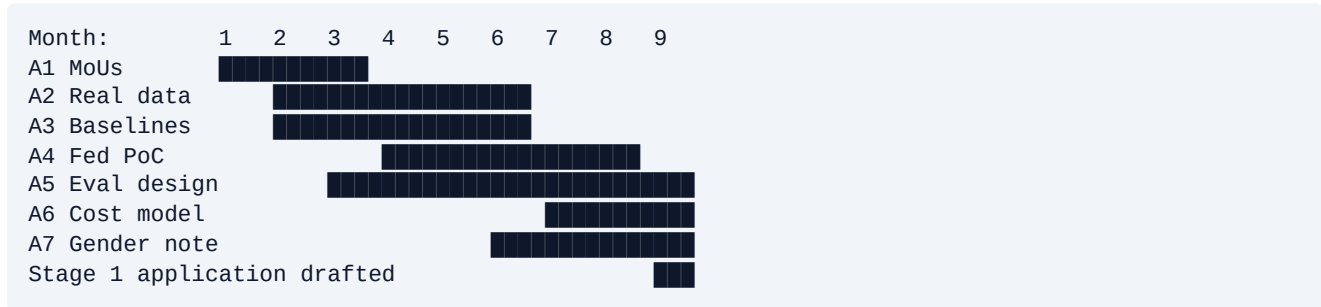
Co-financing: founder's in-kind time and existing MVP codebase (not costed). [TBC: any other co-funding]

10. Risks and mitigations

Risk	Likelihood	Mitigation
SACCOs unwilling to share historical data even under DPPA-compliant terms	Medium	Federated design means raw data never leaves the institution; retrospective validation (A2) can run on-premises with only aggregate results exported. Start with institutions already digitised.
Real fraud typologies differ from simulation; performance drops	Medium	This is precisely what Stage 0 tests; the rule-based MVP provides a floor that works regardless.
High false-positive rate overwhelms small SACCO staff	Medium–High	Threshold tuning on real data; alerts ranked and capped per week; measure investigation burden as an outcome.
Regulatory objection to a shared model hub	Low	Early engagement with UMRA; hub hosted in Uganda; CRB regulations as precedent.
Detection tool changes staff behaviour before evaluation (Hawthorne / deterrence)	Certain	Treat deterrence as a legitimate mechanism; randomisation at institution level captures it; measure behavioural indicators explicitly.

Risk	Likelihood	Mitigation
Applicant entity not eligible	—	Confirm registration before submission; consider a university-led consortium if needed.

11. Timeline (Stage 0)



12. Alignment with FID selection criteria

FID criterion	How this proposal meets it	Gap closed by Stage 0
Impact on people living in poverty	Protects savings and credit access of ~18M mostly rural, low-income SACCO members; prevents institutional collapse that wipes out savings.	Baseline fraud-loss and member-exposure data (A3, A7).
Cost-effectiveness vs. existing approaches	Automated 100% coverage vs. manual audit of 5–10% of loans; marginal cost per institution is small relative to 2–5% annual losses.	Actual unit costs and loss-averted estimates (A6).
Potential for scale and sustainability	Federated hub designed as sector infrastructure (UMRA/UCSCU); open-source integration (Tazama); regional replicability.	Regulator and apex-body engagement (A1).
Innovation	First federated-learning fraud detection for SACCOs/MFIs in Africa; solves data scarcity and privacy-law constraints together.	— (established in the paper)
Readiness for the requested stage	Stage 0 is for promising applications that need preparation before a pilot; the MVP is prototyped and the federated design is published.	Real-data validation (A2, A4).
Rigorous evaluation with a counterfactual	Institution-level cluster RCT with phased roll-out proposed (§5).	Research partner and pre-registered design (A5).
Priority sectors	Not education/health/climate directly; gender equality angle via women's disproportionate reliance on SACCOs — to be substantiated.	Gender analysis (A7).

13. Supporting materials in this repository

- `docs/FedFraudShield_DSA2026_Paper_v3.pdf` — technical paper with architecture, DPPA analysis and simulation results.
 - `mvp/` , `mvp-fast/` — working FraudShield prototype (browser-based; accepts any CSV/Excel export).
 - `backend/` — Python detection engine (rule-based + Isolation Forest / gradient boosting) and smart column-mapping module.
 - `demo financial data/` — synthetic loan datasets used for demonstrations.
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14. Pre-submission checklist

- ☐ Confirm legal entity, registration documents and bank account for FraudShield Uganda
- ☐ Re-read the current FID call guide and rules; confirm Stage 0 ceiling, duration and eligible costs
- ☐ Secure a letter of intent from at least one research partner
- ☐ Secure letters of intent from 3–5 SACCOs/MFIs
- ☐ Obtain a UMRA and/or UCSCU letter of support (or at least a meeting record)
- ☐ Replace every `[TBC]` above
- ☐ Verify all statistics cited (PROFIRA 2021, MTIC sector figures, ACFE 2024) and attach references
- ☐ Prepare 2-page summary and budget in FID's online application format